Mouhamed Nourou Dine Cisse

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PROFESSIONAL SUMMARY

Applied Mathematics and Computer Science engineering student at ENSIIE, pursuing a double degree in Artificial Intelligence (M2IA) with Université Paris-Saclay (2026-2027). Experience in AI, machine learning and data engineering. Completed an academic exchange at McMaster University in Winter 2026.

EDUCATION

ENSIIE and Université Paris-Saclay *2024-2027*

Engineering Degree in Applied Mathematics and Computer Science; double degree, Master 2 in Artificial Intelligence (M2IA), in progress (2026-2027).

M2IA courses (2026-2027): Data Readiness and Evaluation in AI Systems, NLP Today, Deep Learning for NLP, Information Retrieval, Reinforcement Learning, Probabilistic Generative Models, and Signal Processing M2 or Neurosymbolic AI.

McMaster University, Hamilton, Canada *Winter 2026*

Completed academic exchange in Winter 2026; coursework in machine learning, applied AI and generalized linear models.

CPGE MP*, Lycée Albert Schweitzer, Mulhouse *2022-2024*

PROFESSIONAL EXPERIENCE

Applied AI and Data Engineering Intern, Prodigious Groupe, Paris *May-July 2026*

- Built ETL/ELT pipelines from heterogeneous sources to PostgreSQL and prepared data for RAG and LLM use cases.
- Developed internal FastAPI APIs and operational dashboards. Technologies: Python, PostgreSQL, Docker, Azure.

Machine Learning Intern, Bangath Systems and Gainde 2000, Paris and Dakar *May-August 2025*

- Built predictive models on 110,260 Port of Dakar records: Random Forest (0.96 F1-score) and SARIMA (0.67 MAE).
- Preprocessed 136 variables with SQL/PLSQL and deployed FastAPI APIs integrated with Oracle APEX.

SELECTED PROJECTS

Nash-DQN for Multi-Agent Trading, McMaster University *2026*

Implemented a multi-agent trading environment and DQN architectures in PyTorch to study Nash equilibrium learning.

Visual Question Answering (VQA), McMaster University *2026*

Combined ResNet50 image representations and MiniLM embeddings; deployed a BLIP-based visual question answering interface with Streamlit.

Transformer Decoder & LoRA PEFT from Scratch, M2IA *2026*

Built autoregressive decoder (RoPE, SwiGLU, KV cache) and implemented Low-Rank Adaptation (LoRA / PEFT) in PyTorch, reducing trainable parameters by >99.5% with near full fine-tuning performance.

Semantic Representations and Named Entity Recognition, M2IA *2025-2026*

Trained Word2Vec, FastText, CNN, Transformers, and BiLSTM-CNN models for NLP and named entity recognition.

Graph Neural Networks for Malware Detection, Sorbonne Data Challenge *2025*

Classified malware from control-flow graphs using PyTorch Geometric.

TECHNICAL SKILLS

Programming: Python, C++, R, SQL, Java, JavaScript, Bash.

AI and Data: machine learning, deep learning, NLP, computer vision, graph neural networks, reinforcement learning, Transformers, LoRA / PEFT, RAG, ETL/ELT, time series.

Tools: PyTorch, PyTorch Geometric, scikit-learn, Hugging Face, FastAPI, PostgreSQL, Docker, Pandas, NumPy, Git, Linux.

AWARDS AND LANGUAGES

First Prize, ENSIIE Entrepreneurship Challenge (2025), for co-founding ETLia. Ranked first in Data Analysis at ENSIIE.
French: native; English: C1; Arabic: B2; German: B1.